EE656: Robust Control
Assignment 1
Deadline: 30-09-2020

1. Is the state equation

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -3 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u, \qquad y = \begin{bmatrix} 1 & 2 & 1 \end{bmatrix} x$$

   controllable?

2. Is the state equation

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 2 & -1 \end{bmatrix} x + \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix} u, \qquad y = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} x$$

   controllable?

3. Is it true that the rank of $\begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix}$ equals the rank of $\begin{bmatrix} AB & A^2B & \cdots & A^nB \end{bmatrix}$? If not,under what condition will it be true?

4. Is it possible to find a set of b_{ij} such that the state equation

$$\dot{x} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} x + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \\ b_{41} & b_{42} \\ b_{51} & b_{52} \end{bmatrix} u$$

   is controllable?

5. What is the controllability index of the state equation $\dot{x} = Ax + Iu$, where $A \in \mathbb{R}^{\mathtt{n}\times\mathtt{n}}$ and I is $\mathtt{n} \times \mathtt{n}$ identity matrix?

6. Show that the state equation

$$\dot{x} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} x + \begin{bmatrix} B_1 \\ 0 \end{bmatrix} u$$

   is controllable if and only if the pair (A_{22}, A_{21}) is controllable.

7. Consider an uncontrollable system $\dot{x} = Ax + Bu$, with $x \in \mathbb{R}^n$ and $u \in \mathbb{R}^m$. Assume that

$$\text{rank}(\mathcal{C}_{AB}) = \text{rank}\left(\begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix}\right) = r < n.$$

   Let $\{q_1, q_2, ..., q_r\}$ be a basis for the range space of the controllablity matrix $\mathcal{C}_{AB}$, and let $\{q_{r+1}, ..., q_n\}$ be any vectors such that $T := \begin{bmatrix} q_1 & q_2 & \cdots & q_r & q_{r+1} & \cdots & q_n \end{bmatrix}$ is nonsingular. Show that the state transformation

$$x = T\tilde{x} = T \begin{pmatrix} \tilde{x}_c \\ \tilde{x}_{\tilde{c}} \end{pmatrix}, \tilde{x}_c \in \mathbb{R}^r, \tilde{x}_{\tilde{c}} \in \mathbb{R}^{n-r}$$

transforms the given system into the form

$$\begin{pmatrix} \dot{\tilde{x}}_c \\ \dot{\tilde{x}}_{\tilde{c}} \end{pmatrix} = \begin{bmatrix} A_{cc} & A_{c\tilde{c}} \\ 0 & A_{\tilde{c}\tilde{c}} \end{bmatrix} \begin{pmatrix} \tilde{x}_c \\ \tilde{x}_{\tilde{c}} \end{pmatrix} + \begin{bmatrix} B_c \\ 0 \end{bmatrix} u,$$

where (A_{cc}, B_c) is controllable. Show that the uncontrollable modes of the system are given by the set of eigevalues of $A_{\tilde{c}\tilde{c}}$, i.e., $\sigma(A_{\tilde{c}\tilde{c}})$.

8. Show that if (A,B) is uncontrollable, then (A + α I, B) is also uncontrollable for any $\alpha \in \mathbb{R}$.

9. Let λ_i be an eigenvalue of a matrix A and let v_i be a corresponding eigenvector. Let $f(\lambda) = \sum_{k=0}^{l} \alpha_k \lambda^k$ be a polynomial with real coefficients. Show that $f(\lambda_i)$ is an eigenvalue of the matrix function $f(A) = \sum_{k=0}^{l} \alpha_k A^k$. Determine an eigenvector corresponding to $f(\lambda_i)$.

10. For the matrices

$$A_1 = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 5 & 1 & 0 & 0 \\ -1 & 3 & 0 & 0 \\ -1 & -1 & 3 & 0 \\ 1 & 1 & 1 & 4 \end{bmatrix}$$

determine the matrices $A_1^{10}, A_2^{10}, A_3^{10}, e^{A_1 t}, e^{A_2 t}$, and $e^{A_3 t}, t \in R$.

11. Determine some bases for the range and null spaces of the matrices

$$A_1 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}, \; A_2 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}, \text{ and } A_3 = \begin{bmatrix} 3 & 2 & 1 \\ 3 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}.$$

12. Show that for $A = \begin{bmatrix} a & b \\ -b & a \end{bmatrix}$, we have $e^{At} = e^{at} \begin{bmatrix} \cos bt & \sin bt \\ -\sin bt & \cos bt \end{bmatrix}$.

13. Consider the system

$$\dot{x} = Ax + Bu, \; y = Cx$$

with B=0. Let

$$A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \text{ and } C = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}.$$

If possible, select $x(0)$ in such a manner so that $y(t) = te^{-t}, t \geq 0$.

14. Consider the system given by

$$\begin{bmatrix} \dot{x_1} \\ \dot{x_2} \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -\frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{2} \end{bmatrix} u, \; y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

   (a) Determine $x(0)$ so that for $u(t) = e^{-4t}, y(t) = ke^{-4t}$, where k is a real constant. Determine k for the present case. Notice that $y(t)$ does not have any transient components.

   (b) Let $u(t) = e^{\alpha t}$, Determine $x(0)$ that will result in $y(t) = ke^{\alpha t}$, Determine the conditions on α for this to be true. What is k in this case?

15. Consider a system with the following state-space equation:

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ -1 & -2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & -2 & 1 \\ -1 & -1 & -1 & -1 & -3 \end{bmatrix} x + \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ -1 & 1 \\ 0 & 0 \\ 0 & -1 \end{bmatrix} u$$

(a) Prove or disprove the following statement: "There exists an input $u(t)$ such that the system can reach the state $\begin{bmatrix} 2 & -1 & -1 & 0 & 0 \end{bmatrix}$ in finite time".

(b) Design a controller such that the modes of the closed loop system are -1,-2,-3,-4, and -5.

16. Given the present pandemic the faculties have to record their lecture via camera. The idea being floated around is that there is need for a recording system with a camera that follows the faculty when he or she is writing on the blackboard. Hence, there is need of a controller to keep the camera erect when it moves, i.e., the camera needs to be balanced on a stick. The configuration is shown in Figure 1.

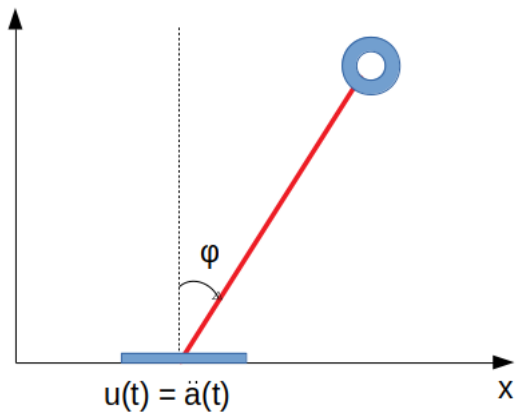

Figure 1: A camera system

The bottom end of the pointer is moving along the x-axis with your input $u(t)$ being the acceleration of this point: $u(t) = \ddot{a}(t)$. The position of the camera is given by $x(t)$. The length of the stick is L; assume that its mass is concentrated at the top end. Assuming ϕ is small the differential equations governing the system are found to be:

$$mg\phi(t) = m\ddot{x}(t) \text{ and } x(t) = a(t) + L\phi(t).$$

(a) Can the system be stabilized by proportional feedback $u(t) = -ky(t)$ (considering $y(t) = \phi(t)$)?

(b) Determine a state-feedback law so that the closed loop system has both the poles at -2.

17. In an alternate universe, you are in-charge of designing the boosters of a satellite to be launched and stationed between earth and the moon. It is known that there is a point between the earth and the moon where the gravitational pull of the earth and the moon cancels each other. This is called the libration point, we call it L. Hence, L is the perfect point for the satellite to be stationed. However, on analysis you realized that if the satellite deviates even the slightest from L, the satellite will either get pulled by earth or the moon. So your task is to design a controller that will keep the satellite at the position

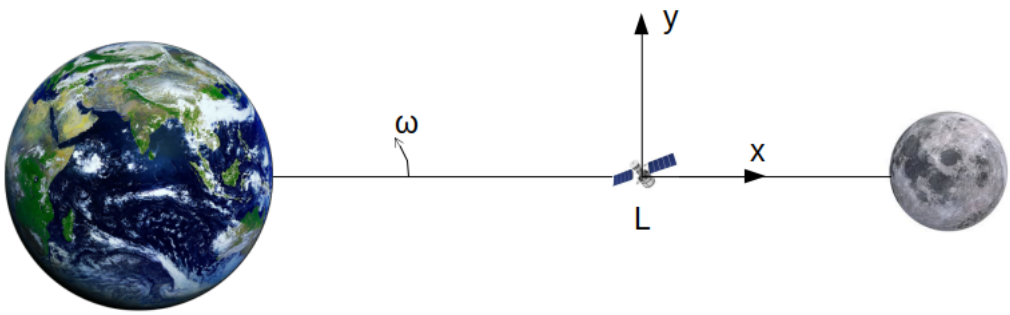

Figure 2: Figure for question 16

L. The dynamic equations for small deviation in position away from the libration point are:

$$\ddot{x} - 2\omega\dot{y} + 9\omega^2 x = 0 \qquad \ddot{y} + 2\omega\dot{x} + 4\omega^2 y = u,$$

where x is the radial position perturbation, y is the azimuthal position perturbation, $u = \frac{F}{m\omega^2}$, F is the engine thrust in the y direction, m is the satellite mass, and $\omega = \frac{2\pi}{29}$ `rad/day`. To stabilize the system you want to use the following state-feedback law

$$u = -k_1 x - k_2\dot{x} - k_3 y - k_4\dot{y}$$

and place the closed-loop system's pole at -3ω, -4ω, and $(-3 \pm 3j)\omega$. Will you be able to design such a controller? If yes, what will be the values of k_1, k_2, k_3 and k_4. If not, why?

18. Consider the platform system shown in Figure 3: it can be used to study suspension systems of automobiles. The system consists of one platform; both ends of the platform

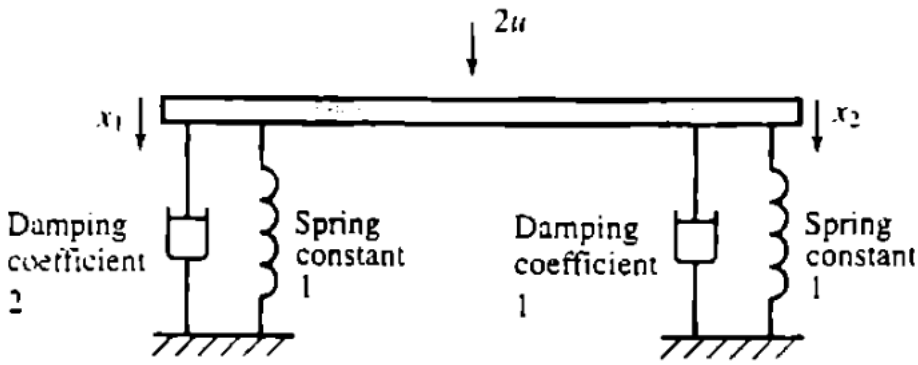

Figure 3: Mass spring system

are supported on the ground by means of springs and dashpots, which provide viscous friction. The mass of the platform is assumed to be zero: thus the movements of the two spring systems are independent and half of the force is applied to each spring system. The spring constants of both springs are assumed to be 1 and the viscous friction coefficients are assumed to be 2 and 1 as shown. If the displacements of the two spring systems from equilibrium are chosen as state variables $x_1(t)$ and $x_2(t)$, then the state equations are $x_1(t) + 2\dot{x}_1(t) = u(t)$ and $x_2(t) + \dot{x}_2(t) = u(t)$.

(a) How much time will it require for the system to reach equilibrium (rest) if no force is applied?

(b) Is it possible to bring the system to rest in 2 `sec` using force? If yes what will be the force that you will apply.

(c) If the viscous friction coefficients and spring constants of both spring systems are all equal to 1, then what will be the force that you would apply to bring the system to rest at 2 `sec`.

19. Consider the network shown in Figure 4. The network has two state variables: the current

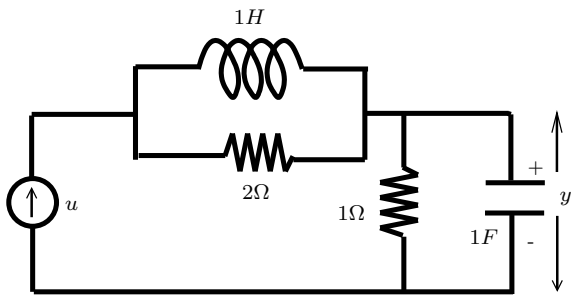

Figure 4: A electrical network

x_1 through the inductor and the voltage x_2 across the capacitor. The output y is measured as the voltage across the capacitor. Find a basis of the subspace of initial conditions that are unobservable. What is the physical interpretation of such a subspace in the context of the circuit in Figure 4.

20. Given a linear time-invariant system $\dot{x} = Ax + Bu$, let

$$\tilde{A} := \begin{bmatrix} A & BB^T \\ 0 & -A^T \end{bmatrix}$$

(a) Verify that $e^{\tilde{A}t}$ has the form

$$e^{\tilde{A}t} = \begin{bmatrix} E_1(t) & E_2(t) \\ 0 & E_3(t) \end{bmatrix}$$

(b) Show that the controllability Gramian of the system is given by

$$W_c(t) = \int_0^t e^{-A\tau} BB^T e^{-A^T\tau} d\tau = E_3^T(t) E_2(t).$$